

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T50 · 3D Pythagoras and Trigonometry

With Answers

Questions with full worked solutions

■ 21 pages

FAMILY MAP

3D Pythagoras Trigonometry

F01

Resolve multi-stage solid geometry

5 Q | 23 marks

F02

Use trigonometry in three dimensions

3 Q | 12 marks

Working Unit 26

4MA1-2H Q24

ORIGINAL QUESTION**4 marks**

24 The diagram shows a square-based pyramid $ABCDE$

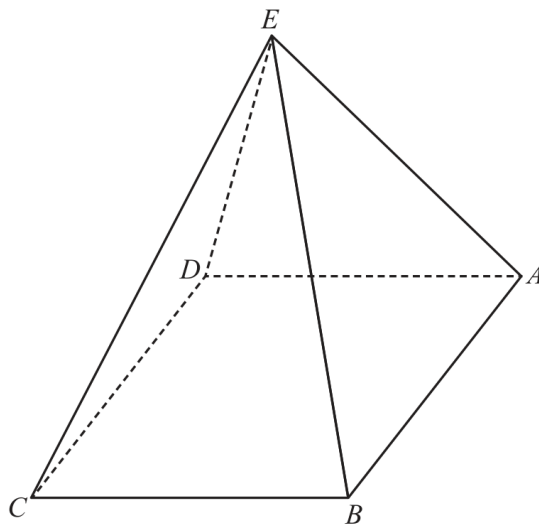


Diagram **NOT**
accurately drawn

$$EA = EB = EC = ED$$

M is the centre of the horizontal square base $ABCD$

Q is the midpoint of AB

Angle $EQM = 80^\circ$

$$EA : AB = n : 1$$

Find the value of n

Give your answer correct to 3 significant figures.

$$n = \dots\dots\dots$$

(Total for Question 24 is 4 marks)

Worked Solution 26

Three-dimensional trigonometry in a pyramid

UNIT 1
4 marks**Choose a scale**

$$\text{Let } AB = 1. \text{ Then } QM = \frac{1}{2} \text{ and } AM = \sqrt{(1/2)^2 + (1/2)^2} = \frac{1}{\sqrt{2}}.$$

Vertical height

$$EM = QM \tan 80^\circ = \frac{1}{2} \tan 80^\circ.$$

Sloping edge

$$EA = \sqrt{EM^2 + AM^2} = \sqrt{\frac{1}{4} \tan^2 80^\circ + \frac{1}{2}} = 2.922474 \dots$$

Final answer

$$n = 2.92 \text{ (3 s.f.)}$$

Original question 17

4MA1-1H Q21

ORIGINAL QUESTION**5 marks**

- 21 The diagram shows the prism $ABCDEFGHJK$ with horizontal base $AEFG$

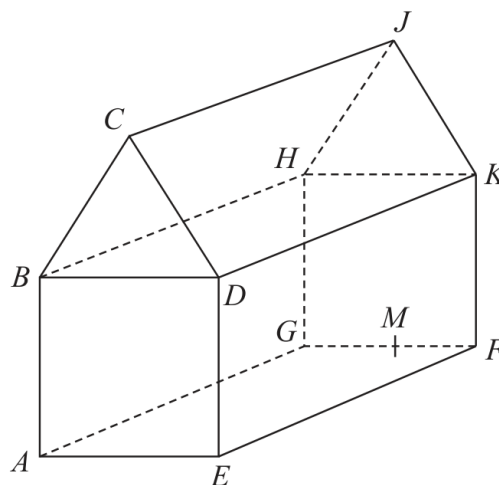


Diagram **NOT**
accurately drawn

$ABCDE$ is a cross section of the prism where
 $ABDE$ is a square
 BCD is an equilateral triangle

$$EF = 2 \times AE$$

M is the midpoint of GF so that JM is vertical.

$$\text{Angle } MAJ = y^\circ$$

$$\text{Given that } \tan y^\circ = T$$

find the value of T , giving your answer in the form $\frac{\sqrt{p} + \sqrt{q}}{17}$ where p and q are integers.

$T = \dots\dots\dots$

Worked solution 17

Chapter: 3D Pythagoras & Trigonometry

MODULAR UNIT 1

5 marks

Q21 – Chapter: 3D Pythagoras & Trigonometry

Family: Resolve multi-stage solid geometry • Variant: Use midpoint or symmetry structure in a 3D model

Method

$$AM = \sqrt{(x^2 + (4x)^2)} = x\sqrt{17}.$$

Why: Use Pythagoras in the horizontal base triangle.

Method

$$\text{The equilateral-triangle height is } \sqrt{((2x)^2 - x^2)} = x\sqrt{3}.$$

Why: Use the altitude of the equilateral cross-section.

Method

$$\tan y = (x\sqrt{3} + 2x) / (x\sqrt{17}) = (\sqrt{3} + 2) / \sqrt{17}.$$

Why: Use opposite over adjacent for angle MAJ.

Method

$$\text{Rationalising gives } (\sqrt{3} + 2)\sqrt{17}/17 = (\sqrt{51} + 2\sqrt{17})/17.$$

Why: Multiply numerator and denominator by $\sqrt{17}$.

Method

$$\text{Since } 2\sqrt{17} = \sqrt{68}, T = (\sqrt{51} + \sqrt{68})/17.$$

Why: Write the answer in the requested surd form.

Final answer

$$T: (\sqrt{51} + \sqrt{68})/17$$

19 The diagram shows a cuboid $ABCDEFGH$.

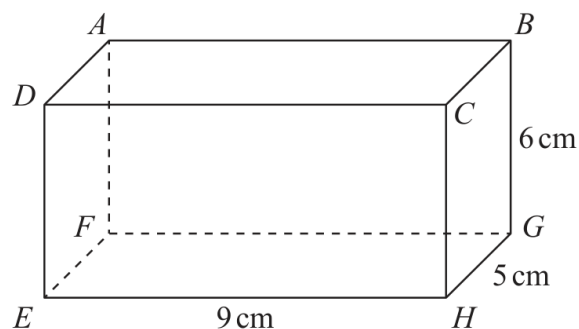


Diagram **NOT**
accurately drawn

$EH = 9\text{ cm}$, $HG = 5\text{ cm}$ and $GB = 6\text{ cm}$.

Work out the size of the angle between AH and the plane $EFGH$.
Give your answer correct to 3 significant figures.

o

Worked solution

January 2020 | Question 19 | 4 marks

Family: Use trigonometry in three dimensions • Variant: Find angle between a line and a plane

Method / working

1. Horizontal distance AH projection is $\sqrt{9^2+5^2}=\sqrt{106}$.
2. Vertical height is 6 cm.
3. $\tan\theta=6/\sqrt{106}$.
4. $\theta=30.2^\circ$ to 3 s.f.

Final answer: 30.2°

- (b) Calculate the size of the angle between AF and the plane $ADEH$.
Give your answer correct to one decimal place.

.....
(3)

Worked solution

January 2021 | Question 17 | 3 marks

Family: Use trigonometry in three dimensions • Variant: Find angle between a line and a plane

Method / working

1. Horizontal distance to the point below F is $\sqrt{24^2 + 32.5^2} = 40.4$ m.

2. $AF = \sqrt{40.4^2 + 20^2} = 45.08$ m.

3. $\sin \theta = 20/45.08$, giving $\theta = 26.3^\circ$ to 1 d.p.

Final answer: 26.3°

- 22 The diagram shows a triangular prism $ABCDEF$ with a horizontal base $ABEF$.

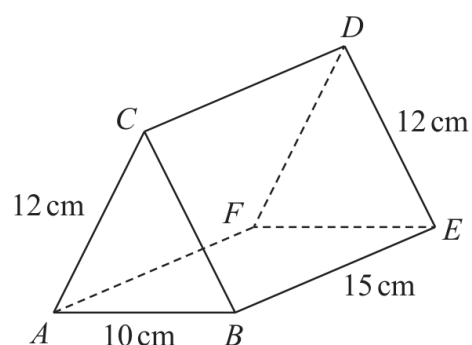


Diagram **NOT**
accurately drawn

$$AC = BC = FD = ED = 12 \text{ cm} \quad AB = 10 \text{ cm} \quad BE = 15 \text{ cm}$$

Calculate the size of the angle between AD and the base $ABEF$.
Give your answer correct to 3 significant figures.

..... °

Worked solution

June 2021 | Question 22 | 4 marks

Family: Resolve multi-stage solid geometry • Variant: Use a non-rectangular base before a line-plane angle

Method / working

1. Use the prism diagram to form the right triangles with lengths 15.8, 19.2 and 10.9 cm.
2. Use $\tan(\text{angle}) = 10.9/15.8$ (equivalently a sine/cosine relation).
3. Evaluate the inverse trigonometric function.
4. Round to 1 decimal place: 34.6° .

Final answer: 34.6°

18 The diagram shows cuboid $ABCDEFGH$.

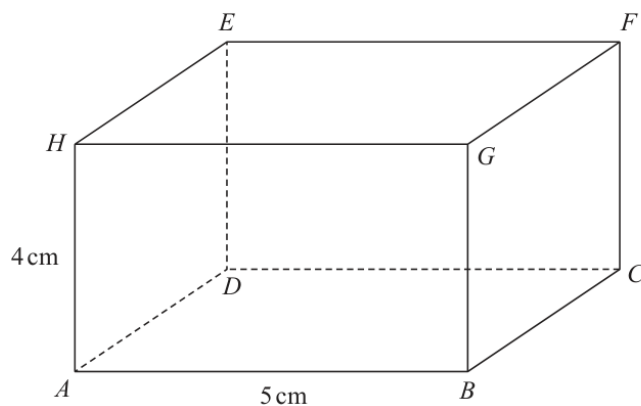


Diagram **NOT**
accurately drawn

$$AB = 5 \text{ cm}$$

$$AH = 4 \text{ cm}$$

The size of the angle between CH and the plane $ABCD$ is 35°

Calculate the volume of the cuboid.

Give your answer correct to 3 significant figures.

.....cm³

(Total for Question 18 is 5 marks)

Answer reference | May-Nov 2020 | Question 18

Family: Use trigonometry in three dimensions • Variant: Find angle between a line and a plane

18	eg $\frac{4}{AC} = \tan 35$ oe or $\frac{AC}{4} = \tan 55$ oe or $\frac{AC}{\sin 55} = \frac{4}{\sin 35}$ oe or $CH = \frac{4}{\sin 35}$ oe (= 6.97...) and $\frac{AC}{6.97} = \cos 35$ oe or $CH = \frac{4}{\sin 35}$ oe (=6.97...) and $AC^2 = 6.97^2 - 4^2$ oe			M1	A correct trig statement involving AC or trig and then Pythagoras involving AC
	$(AC =) \frac{4}{\tan 35}$ oe eg $(AC =) 4 \tan 55$ (= 5.71...) or $(AC =) \frac{4 \sin 55}{\sin 35}$ or "6.97" $\times \cos 35$ oe or $(AC =) \sqrt{6.97^2 - 4^2}$			M1	complete method to find AC
	$(BC =) \sqrt{5.71^2 - 5^2}$ (= 2.76...)			M1	complete method to find BC
	$4 \times 5 \times "2.76..."$			M1	method to find volume
		55.3	5	A1	accept 55.1 – 55.5
					Total 5 marks

Worked solution

May-Nov 2020 | Question 18 | 5 marks

Family: Use trigonometry in three dimensions • Variant: Find angle between a line and a plane

Official final mark-scheme pages are embedded immediately after this question.

23 The diagram shows a solid prism $ABCDEFGHJ$

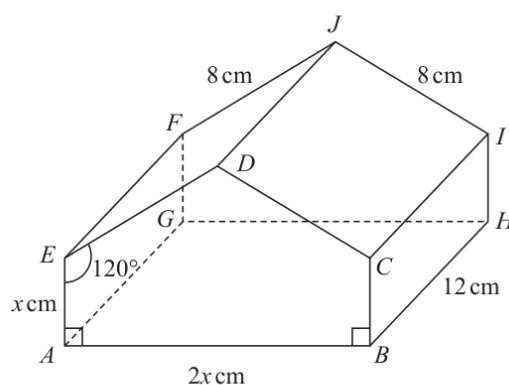


Diagram **NOT**
accurately drawn

The prism is such that each cross section is a pentagon where

$$AE = BC = x \text{ cm} \quad AB = 2x \text{ cm} \quad ED = CD = 8 \text{ cm}$$

$$\text{angle } EAB = \text{angle } CBA = 90^\circ \quad \text{angle } AED = \text{angle } BCD = 120^\circ$$

Given that $AG = BH = EF = DJ = CI = 12 \text{ cm}$

calculate the angle that AJ makes with the base $ABHG$ of the prism.

Give your answer correct to 3 significant figures.



(Total for Question 23 is 5 marks)

Answer reference | January 2023 | Question 23

Family: Resolve multi-stage solid geometry • Variant: Use 3D trigonometry to obtain a terminal volume

Q	working	Answer	Mark	Notes
23	$[DN =]8\sin 30 \text{ or } 8\cos 60 (= 4) \text{ oe [where } N \text{ is the midpoint of } EC]$ or $[x =]8\cos 30 \text{ or } 8\sin 60 (= 4\sqrt{3} = 6.928\dots)$ or $2x = \sqrt{8^2 + 8^2 - 2 \times 8 \times 8 \times \cos 120} (= \sqrt{192} = 8\sqrt{3} = 13.85\dots)$		5	M1
	$[DN =]8\sin 30 \text{ or } 8\cos 60 (= 4) \text{ oe eg } \sqrt{8^2 - (4\sqrt{3})^2} (= 4)$ And 1 of $[x =]8\cos 30 \text{ or } 8\sin 60 (= 4\sqrt{3} = 6.928\dots) \text{ oe or}$ $\sqrt{8^2 - "4"^{n2}} (= 4\sqrt{3} = 6.928\dots) \text{ or}$ $2x = \sqrt{8^2 + 8^2 - 2 \times 8 \times 8 \times \cos 120} (= 8\sqrt{3} = 13.85\dots)$			M1 [(JM =) 4 + 4\sqrt{3} implies M2]
	$[AM =]\sqrt{12^2 + ("4\sqrt{3}")^2} (= \sqrt{192} = 8\sqrt{3} = 13.856\dots) \text{ oe}$ [where M is the midpoint of GH]			M1 Clear intention to be AM (not EC)
	$\tan MAJ = \left(\frac{"4" + "4\sqrt{3}"}{8\sqrt{3}} \right) \text{ oe eg } \tan MAJ = \left(\frac{10.928\dots}{13.856\dots} \right)$ if student uses sin or cos, then $AJ = 17.647\dots$ to award marks this must come from a correct method or be correct			M1
	Correct answer scores full marks (unless from obvious incorrect working)	38.3		A1 Accept 38.1 – 38.4 If no marks scored then award SCB2 for $\tan MAJ = \frac{4+x}{\sqrt{x^2 + 12^2}}$ or SCB1 for $AM = \sqrt{x^2 + 12^2}$
Total 5 marks				

Worked solution

January 2023 | Question 23 | 5 marks

Family: Resolve multi-stage solid geometry • Variant: Use 3D trigonometry to obtain a terminal volume

Official final mark-scheme pages are embedded immediately after this question.

26 Here is a prism $ABCDEFGH$ with a horizontal, rectangular base $ABGF$

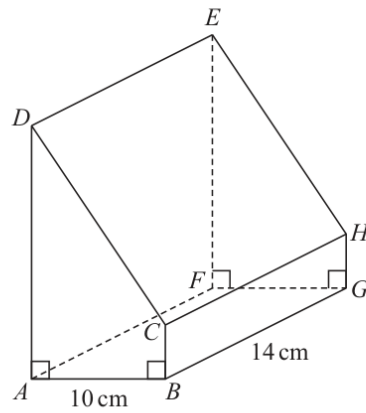


Diagram **NOT**
accurately drawn

$$AB = 10 \text{ cm} \qquad BG = 14 \text{ cm}$$

angle DAB = angle ABC = angle EFG = angle FGH = 90°

$$BC = \frac{1}{5}AD$$

The angle of elevation of E from B is 50°

Calculate the volume of the prism.

Give your answer correct to 3 significant figures.

Show your working clearly.

..... cm³

(Total for Question 26 is 5 marks)

Worked solution

November 2025 | Question 26 | 5 marks

Family: Resolve multi-stage solid geometry | Variant: Use 3D trigonometry to obtain a terminal volume

METHOD / WORKING

Worked solution
November 2025 Question 26 5 marks
Family: Resolve multi-stage solid geometry Variant: Use 3D trigonometry to obtain a terminal volume
Method
The horizontal base ABGF is a rectangle.
$AB=10, \text{ and } BG=14$
So the horizontal distance from B to F is
$BF=\sqrt{10^2+14^2}=\sqrt{296}$
The angle of elevation of E from B is 50° .
Since E is vertically above F,
$\tan 50^\circ = \frac{FE}{BF}$
$FE=\sqrt{296}\tan 50^\circ$
The prism has the same cross section throughout, so
$AD=FE=\sqrt{296}\tan 50^\circ$
Also
$BC=\frac{1}{5}AD$
Area of the trapezium cross section ABCD:
$\frac{1}{2}(AD+BC) \times AB$
$=\frac{1}{2}(AD+\frac{1}{5}AD) \times 10$
$=6AD$
Volume of the prism:
$6AD \times 14=84AD$
$=84\sqrt{296}\tan 50^\circ$
$=1722.311\dots$
Correct to 3 significant figures:
1720
Answer: $\boxed{1720\text{cm}^3}$
PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION

FINAL ANSWER